

General Software Platform for Designing and Developing of Augmented Reality Task Support Systems

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ABSTRACT

Augmented reality (AR) task support systems have not yet been widely spread in actual companies despite the fact that many researchers have worked tirelessly to validate their effectiveness for various industrial tasks. One of the reasons for this is the excessive development cost of a practical AR system. Development still strongly depends on outstanding developers' skills and trial-and-error. To solve this problem, we propose to separate an AR task support system into a reusable software part (including basic functions and its user interfaces) and task support contents that differ depending on each task. We have developed a general software platform and an original file format for describing task support contents based on analyzing various tasks. In addition, we have also proposed a web tool that enables easy design of AR task support content even for those without AR development experience. The output of the tool, an XML file written in the format, is loaded into the software platform to enable operators to use the AR task support system. We demonstrate a series of steps from the creation of task support content to the use of the AR task support system.

ACM Reference Format:

Soshiro Ueda, Keishi Tainaka, Yiming Shen, Shuntaro Ueda, Konstantin Kulik, Yuichiro Fujimoto, Taishi Sawabe, Masayuki Kanbara, and Hirokazu Kato. 2023. General Software Platform for Designing and Developing of Augmented Reality Task Support Systems. In *SIGGRAPH Asia 2023 XR (SA XR '23)*, December 12–15, 2023, Sydney, NSW, Australia. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3610549.3614599>

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SA XR '23, December 12–15, 2023, Sydney, NSW, Australia

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ACM ISBN 979-8-4007-0316-4/23/12.

<https://doi.org/10.1145/3610549.3614599>

1 INTRODUCTION

In the decades, researchers have demonstrated the effectiveness of augmented reality (AR) task support systems in a wide variety of tasks, especially industrial tasks involving assembly, and inspection. A couple of large companies introduced Microsoft HoloLens2 for actual tasks, however, it is still difficult to say that many companies have introduced it to their workplaces. One of the reasons for this is the large development cost of a practical AR task support system for each use case. Furthermore, the current development depends on the skills of experienced developers and trial-and-error at the work site, which requires a large time and financial costs to achieve the practical effect.

In a series of studies, we have proposed separating an AR task support system into a reusable software platform including basic functions and corresponding user interface (UI) and task support contents that differ depending on each task. Minimal development for only the latter will result in significant cost savings. We analyzed existing AR task support systems and other software with a separation structure between platforms and contents to develop a general software platform [Shen et al. 2023]. The dedicated XML file format was specified for describing task support content based on analyzing various tasks. Based on these, we have also developed a web tool that enables users without sufficient experience in AR development to easily design and develop AR task support content [Tainaka et al. 2023]. The tool outputs an XML file described in the above format that contains the work process, the URLs to visual assets (i.e., images, videos, 3D models), and their visualization method. The AR task support system is completed by just loading this file into the software platform on a head-mounted display (HMD) and localizing each asset to a specific location. This demonstration shows how easy it is to build a high-quality task support system through the above sequence of steps.

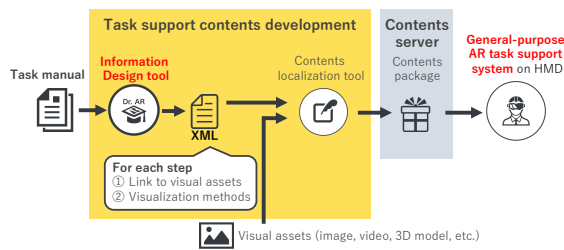


Figure 1: Flow of the development of task support contents.

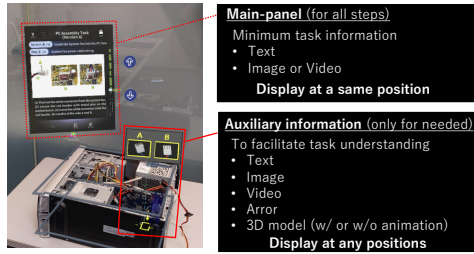


Figure 2: Main-panel and sub-panels (auxiliary information)

2 GENERAL SOFTWARE PLATFORM

2.1 Overview

Figure 1 shows the flow of the development and the use of task support contents based on our method. Here, we assume that “a task” is composed of multiple “steps” in a sequential manner basically. We also define the person who creates task support content as “a developer” and the person who actually works with the AR system as “an operator”. First, information on each step is entered into a dedicated web tool [Tainaka et al. 2023], based on the original procedure manual of the target task by the developer. According to the tool’s instructions, the developer determines the characteristics of each step and the links to the visual assets data. The developer also decides on a specific method for how to visualize each of the visual assets to operators so that they can efficiently understand and proceed with a step (“visualization method”). The tool outputs a summary file of the determined information according to the dedicated format [Shen et al. 2023]. Next, the content localization tool on AR-HMD reads the file. The developer manually places the visual assets in the real environment, making a final content package including all the information. This content package is uploaded to a dedicated server. Operators can utilize any task support content by accessing this server using a general-purpose software platform running on the HMD and downloading the package.

2.2 Design Policy in Task Support Content

The supporting information used in the General Software Platform can be classified into the main panel, which is always displayed in every step with a consistent appearance, and auxiliary information (sub-panels, and 3D models), which is used only in necessary steps (Figure 2). The main panel consists of the current step information (name and ID), instructional text, and an image /short video indicating the work to be done. Our main policy for task support content design is not to increase the operator’s cognitive load by providing excessive assistance. We believe that the main panel is sufficient for the operator to understand the work in most steps and

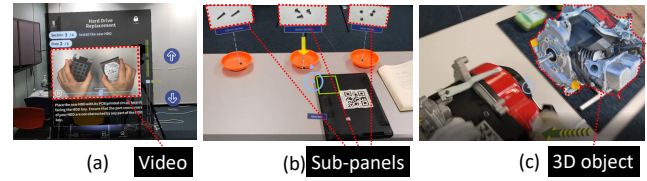


Figure 3: Example of constructed contents. (a) A main panel with video displayed. (b) Multiple sub-panels showing storage locations for different screws. (c) A 3D model of the engine displayed next to the actual engine, showing the current step.

more sophisticated visualization methods should only be used for necessary steps. For some complicated steps, flexible task support is possible by combining auxiliary information (See Figure 3).

2.3 Web Tool for Designing Contents

We have proposed Dr.AR (<https://ar-guideline.naist.jp/?lang=en>), a web tool for designing task support content using AR. This tool aims to make it easy for developers with no/little experience in AR development to create a highly effective AR task support system. The developer fills in the name of each step and the text to be presented to the operator. Next, the developer classifies each step into one of the subtask categories [Tainaka et al. 2023]. This is because the available visualization methods differ according to each subtask category. As of June 2023, our tool provides a total of 30 different visualization methods. In addition, by specifying the characteristics of each step (e.g., “Need the accuracy within 1 cm”) and development requirements (e.g., no 3D model available) via checkboxes, the set of recommended methods is displayed to the developer. Selecting a preferred method among recommended methods will be exported to the file.

3 DEMONSTRATION

In the demonstration, visitors will be able to briefly experience a series of creation of task support content and task performing with the created content on HoloLens2. First, visitors freely enter the contents of one step of a given task on a laptop using a web tool, and select a visualization method (using prepared images and 3D models in advance). The information is immediately uploaded to the server. The visitor then puts on Microsoft HoloLens2, downloads the created content from the server, and can immediately experience it as an all-in-one AR task support system. In addition, we plan to have the visitors experience flexible task support methods through the proposed framework using the contents prepared in advance for the specific task. Figure 3 shows examples of constructed contents.

ACKNOWLEDGMENTS

This work was partially supported by JSPS KAKENHI Grant Number JP22K12127.

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